

L1.8 In-class Assignment – Lists and Data Frames

CPSC 292 – 9/11/2026

Name: _____ Section: _____

Collaboration team: _____

Question 1

Describe in which ways accessing items in lists and data frames are different and in which ways they are similar.

Question 2

Describe the ways in which data frames are similar to matrices and in which ways they are different.

Question 3

Construct a list that has five members, each member is a vector of length 10 of a different atomic data type in R.

Question 4

Replace at least one word in the character data type of your list in Question 3 with the word “dingbat” using `sub`.

Question 5

▼ penguins	344 obs. of 8 variables	
\$ species	: Factor w/ 3 levels "Ade..."	
\$ island	: Factor w/ 3 levels "Bis..."	
\$ bill_len	: num 39.1 39.5 40.3 NA ...	
\$ bill_dep	: num 18.7 17.4 18 NA 19...	
\$ flipper_len	: int 181 186 195 NA 193...	
\$ body_mass	: int 3750 3800 3250 NA ...	
\$ sex	: Factor w/ 2 levels "fem..."	
\$ year	: int 2007 2007 2007 200...	

Figure 1: The penguins data set

- For the `penguins` data set in Fig. 1, describe the data in each column and list the class.
- You would like to add a column of data to the `penguins` data set (Fig. 1) describing how tall the penguins are. Write a line of code that adds the column assuming the height of each penguin in the data set is stored in a vector named `height`.
- You need to calculate the mean body mass of female penguins in this data set. Write a line of code that does this. (Assume that female penguins are coded as `female`.)